

Scientific Retractions by Brazilian Authors in Engineering: A Descriptive Study Using the Retraction Watch/Crossref Database

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Abstract—Retractions constitute a fundamental self-correction mechanism of scientific literature, yet their occurrence in emerging fields such as biomedical engineering (BME) remains insufficiently characterized in the Brazilian context. This study presents a descriptive survey of retractions involving Brazilian-affiliated authors in engineering using the Retraction Watch/Crossref database (71,106 records, accessed July 2026). No retractions specifically classified as BME with Brazilian authorship were identified. A result consistent with prior Latin American studies points to Internal Medicine, Oncology, and Health Sciences in general as the predominant domains of retraction in the region, in contrast to more recent, lower-volume engineering fields. The scope was consequently broadened to all engineering subfields indexed in the database, yielding 29 cases, classified into three categories: fraud/error/data issues ($n = 10$), plagiarism/duplication ($n = 9$), and other reasons ($n = 10$). Findings are discussed considering the Latin American research-integrity literature, with implications for the Brazilian BME research community.

Keywords— scientific retraction; research integrity; Biomedical Engineering; scientific misconduct; Retraction Watch; Latin America.

I. INTRODUCTION

The retraction of a scientific article is the formal mechanism by which journals and authors indicate that a publication should no longer be considered a valid part of the scientific record, whether due to an unintentional error or deliberate misconduct (fabrication, falsification, or plagiarism) [2–4]. The absolute number of retractions has grown substantially over the past two decades, a phenomenon attributed, in non-mutually exclusive fashion, to a genuine increase in misconduct, to enhanced editorial oversight, and to the widespread adoption of automated tools for detecting plagiarism and duplication [3,4].

Evidence from large-scale studies indicates that misconduct, rather than genuine error, accounts for the majority of biomedical retractions: a review of 2,047 retracted biomedical articles found that 67.4% were attributed to misconduct (fraud, duplication, or plagiarism) compared to only 21.3% attributed to genuine error [4]. Complementarily, estimates derived from anonymous surveys suggest that about 2% of researchers self-report having fabricated or falsified data at least once, and up to 34% admit to other questionable research practices [2].

In the Latin American context, particularly in Brazil, this phenomenon has received increasing attention in the scientific integrity literature, albeit more recently than in countries with a more established scientific tradition [5–7]. Biomedical Engineering (BME), as a comparatively recent discipline with a lower volume of research output than Medicine or the Health Sciences broadly construed, has yet to be systematically characterized with respect to its exposure to retractions. The present study aims to characterize retractions by authors with Brazilian affiliations in engineering, using the Retraction Watch/Crossref database, with an initial focus on BME and, given the absence of specific cases, an extension to the set of engineering subfields indexed in the database.

II. MATERIALS AND METHODS

A. Data Source

We used the complete Retraction Watch Database (retraction_watch.csv file, 71,106 records, version accessed in July 2026), covering retractions recorded between July 15, 2008 (first case) and November 6, 2025 (last case), made publicly available by Crossref since its acquisition of the database in 2023 [1]. It is currently the most comprehensive publicly available retraction database, aggregating notifications sourced from publishers' websites, bibliographic databases, and communications from the scientific community, and updated on business days.

B. Filtering Criteria

The database was initially filtered by the "Country" field containing "Brazil," resulting in 488 records. This subset was then further filtered by cross-referencing the "Subject", "Institution", and "Title" fields with the terms "biomedical engineering", "bioengenharia", and "engenharia biomédica.". No record in this filtered set corresponded to a retraction genuinely attributable to the field of a biomedical engineering (BME) article with a Brazilian authorship. Notably, the database's controlled subject vocabulary does not include a "Biomedical Engineering" category but only general engineering subfields (Chemical, Electrical, Structural, Mechanical, and General). The only textual occurrences of "biomedical" terms were associated with the Institute of Biomedical Sciences at the University of São Paulo (USP), a unit focused on basic biomedical sciences rather than engineering, and were therefore excluded from the analysis.

C. Scope Extension

Given the absence of cases in this specific area, a finding consistent with previous research on the profile of Latin American retractions [5–7], the search criteria were expanded to include any subfield of engineering classified in the database, while retaining the requirement for Brazilian affiliation, resulting in 29 records.

D. Classification

Each of the 29 records was classified into one of three categories based on the content of the “Reason” field, which in the database is populated using a controlled vocabulary, sometimes with multiple reasons per record: (i) fraud, error, or data issues (falsification/fabrication of data or images, unreliable data or results, image manipulation); (ii) plagiarism or duplication (duplication of an article, text, or image; plagiarism); and (iii) other reasons (compromised peer review, disputes over authorship/affiliation, copyright claims, administrative errors by the journal). When a record cited reasons spanning more than one category, the category of greatest severity (fraud > plagiarism/duplication > other reasons) was assigned as the primary classification.

E. Methodological Limitations

The approach adopted depends entirely on the coverage and granularity of Retraction Watch’s subject vocabulary, which, as noted, does not distinguish biomedical engineering from other subfields of engineering. This implies that retractions occasionally published in indexed journals under generic labels (e.g., “Medicine-Biomedical” or “Engineering-General”) may not have been captured by the text filters employed, even if they contain Biomedical Engineering content. The supplementary consultation of national databases such as SciELO and LILACS, in which Almeida et al. [5] identified a high prevalence of plagiarism-related retractions represents a priority direction for future studies seeking broader coverage of the Brazilian research landscape.

III. RESULTS

Table I summarizes the distribution of the 29 retractions by Brazilian authors in engineering according to the predominant category of the reason for retraction, and Table II details their distribution by engineering subfield (a single record may be associated with more than one subfield).

TABLE I
DISTRIBUTION OF RETRACTIONS BY CATEGORY

Category	N
Fraud, error or data	10
Plagiarism or duplication	9
Other	10
Total	29

TABLE II
DISTRIBUTION BY ENGINEERING SUBFIELD

Engineering Subfield	N
Chemical	11
Structural	7
Electrical	5
Mechanical	3
General	3

The category “fraud, error, or data” included cases of unreliable data or results, image manipulation, and data falsification or fabrication, many of which were identified through investigations conducted by the journal itself or by third parties. A definitive association with industrial-scale fabrication schemes (“paper mills”) could not be established based on the fields available in the database. The “plagiarism or duplication” category primarily consisted of duplication of articles, text, or images, including cases attributable to editorial errors rather than to author misconduct. The “other” category mainly comprised cases of compromised peer review (fraud in the editorial process, not in the scientific content itself), disputes over authorship or affiliation, and copyright claims.

Regarding the distribution by subfield, a concentration was observed in Chemical Engineering (11 cases) and Structural Engineering (7 cases), subfields with the highest historical publication volume in Brazil—followed by Electrical Engineering (5), Mechanical Engineering (3), and General Engineering (3). No cases were identified with biomedical engineering, including biomedical instrumentation, biomaterials for medical devices, and biomedical signal processing.

IV. DISCUSSION AND CONCLUSION

The absence of Brazilian retractions specifically in Biomedical Engineering is consistent with the field’s still relatively recent emergence and lower volume of output in Brazil compared to more traditional medical and engineering fields, and is also in line with the findings of previous Latin American studies, which identify a concentration of retractions in the region in medical and health sciences journals [5–7]. This does not mean, however, that there is no risk: Brazilian biomedical engineering researchers frequently publish in international journals of medicine, bioengineering, or combined electrical/biomedical engineering, which can hinder their identification in databases employing subject vocabularies with low granularity, as evidenced by the very absence of a “Biomedical Engineering” category in Retraction Watch’s controlled vocabulary.

The concentration of retractions in Brazilian chemical and structural engineering, and the predominance of reasons linked to compromised peer review and duplication/editorial errors (and not necessarily deliberate fraud by the author) in the sample, suggests that systemic editorial vulnerabilities, and not solely individual misconduct, play a significant role, a pattern also discussed by Almeida et al. [5] for the SciELO/LILACS databases. The absence of specific cases of research misconduct does not indicate an absence of risk, but rather possible underreporting: publications in this field often appear in hybrid journals (Electrical Engineering, Medicine, International Bioengineering), which makes it difficult to track them under a specific label in the database. Furthermore, the observed pattern—a concentration in fields with higher publication volumes and a predominance of editorial errors over deliberate fraud—suggests that exposure to retractions tends to increase alongside the volume of scientific output. Although the data presented here do not allow for the projection of future trends, the trajectory observed in fields with higher publication volumes suggests, by analogy, that the growth of scientific output in bioengineering tends to

coincide with greater exposure to retractions. In this regard, and as a recommendation drawn from the scientific integrity literature [5–7], the proactive reinforcement strengthening of peer review and data management practices within the Brazilian BME community is desirable.

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CONFLICTS OF INTEREST

The authors declare that they have no conflict of interest relative to the present work.

REFERENCES

- [1] Retraction Watch Database / Crossref. Available at: <https://retractiondatabase.org> e <https://gitlab.com/crossref/retraction-watch-data>. Acesso em 2026.
- [2] D. Fanelli, "How many scientists fabricate and falsify research? A systematic review and meta-analysis of survey data," *PLoS ONE*, vol. 4, no. 5, e5738, 2009.
- [3] R. G. Steen, "Retractions in the scientific literature: is the incidence of research fraud increasing?" *J. Med. Ethics*, vol. 37, no. 4, pp. 249-253, 2011.
- [4] F. C. Fang, R. G. Steen, and A. Casadevall, "Misconduct accounts for the majority of retracted scientific publications," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 109, no. 42, pp. 17028-17033, 2012.
- [5] R. M. V. R. Almeida, K. A. Rocha, F. Catelani, A. J. Fontes-Pereira, and S. M. R. Vasconcelos, "Plagiarism allegations account for most retractions in major Latin American/Caribbean databases," *Sci. Eng. Ethics*, vol. 22, no. 5, pp. 1447-1456, 2016.
- [6] C. Candal-Pedreira et al., "Evolution and characterization of health sciences paper retractions in Brazil and Portugal," *Account. Res.*, vol. 30, no. 8, pp. 725-742, 2023.
- [7] P. Herrera-Añazco et al., "Retraction of health science articles by researchers in Latin America and the Caribbean: A scoping review," *Developing World Bioethics*, vol. 25, no. 1, pp. 5-15, 2025.